

8 Select a number

Now use the `randrange()` function from the `random` module to select a random number from 0 to 99. Put this line at the bottom of your code.

```
noun = random.choice(nouns)
number = random.randrange(0, 100)
```

9 Select a special character

Using the `random.choice()` function again, add this line to pick a random punctuation character. This will make your password even harder to crack!

```
number = random.randrange(0, 100)
special_char = random.choice(string.punctuation)
```

This is a constant.

10 Create the new secure password

It's time to assemble all the different parts to create the new secure password. Type these two lines of code at the end of your program.

```
password = adjective + noun + str(number) + special_char
print('Your new password is: %s' % password)
```

Your secure password will be stored in this variable.

This changes the random number into a string.

This displays the new password in the shell.

EXPERT TIPS

Constants

A constant is a special type of variable whose contents can't be changed. The constant `string.punctuation` holds a string of characters used for punctuation. To see what it holds, type `import string` into the shell, followed by `print(string.punctuation)`.

```
>>> import string
>>> print(string.punctuation)
! " # $ % & ' ( ) * + , - . / : ; < = > ? @ [ \ ] ^ _ ` { | } ~
```

Characters in this constant

EXPERT TIPS

Strings and integers

The `str()` function turns a whole number (an integer) into a string. If you don't use this function, Python shows an error when you try to add an integer to a string. Test it: type `print('route '+66)` into the shell window.

To avoid this error, use the `str()` function to change the number into a string first.

```
>>> print('route '+66)
```

Traceback (most recent call last):

File '<pyshell#0>', line 1, in <module>

```
print('route '+66)
```

TypeError: Can't convert 'int' object to str implicitly

Error message

```
>>> print('route '+str(66))
route 66
```

The number goes inside the brackets of the `str()` function.